

ER Physiology Practice Questions with answers

Topic: Thyroid gland

1. A 28-year-old man has decided to fast for 48 hours for charity. He is concerned and asks you if he needs to worry about the lack of iodine in his diet during this time causing decreased synthesis of thyroid hormones. Which of the following statements would apply in this case?

- A. There is no deficiency as he has a large store of hormones in the thyroid gland.
- B. He would be deficient as continuous supply of iodine is needed
- C. No deficiency as there is large iodine store in your bones
- D. There is deficiency as the liver breaks down all iodine
- E. No deficiency as they have a large iodine store bound to plasma proteins

2. A 21-year-old woman comes the physician because of a 3-month history of 4.53-kg (10-lb) weight gain and difficulty coping with the winter weather. He says that he always feels cold even on a hot day. Physical examination shows puffy face and a 1-cm size nodule palpable in her neck. Blood studies show:

TSH level: High

T4 level: Low

TSH-R-AB [stim]: Negative.

Which of the following best describes this patient's condition?

- A. Grave's Disease
- B. Secondary hypothyroidism
- C. Primary hyperthyroidism
- D. Iodine deficiency
- E. Secondary hyperthyroidism

3. Under normal circumstances, which of the following structures concentrates the highest amount of iodine?

- A. Extracellular fluid
- B. Adrenal gland
- C. Thyroid gland
- D. Liver
- E. Kidney

Answer Key: C.

Most of the iodine is concentrated in the thyroid gland in the colloid in the thyroglobulin molecule.

4. A 22-year-old woman comes to the physician for a follow up examination. She was diagnosed with hypothyroidism 2 years ago. Physician orders for thyroid function tests. Laboratory studies show a low T3 & T4 and elevated levels of TRH & TSH. Which of the following is the most likely mechanism of action of TRH in the thyrotrophic cells of the anterior pituitary?

- A. IP3 and DAG second messenger system
- B. cAMP second messenger system
- C. Depolarizing the cell
- D. Binding to receptors in the nucleus of the cell
- E. Activating tyrosine kinase

5. A 34-year-old woman comes to the physician because of a 6-week history of heat intolerance, nervousness, increased appetite, and weight loss. Physical examination shows warm and moist skin, stare look, fine tremors of hands, and hyperreflexia. Laboratory studies show elevated serum concentrations of T3 and T4. Which of the following is the most likely mechanism of action of these hormones in the target cells?

- A. The IP3 and DAG second messenger system
- B. cAMP second messenger system
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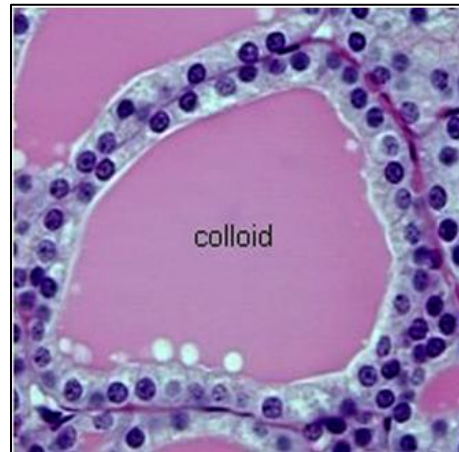
6. A 25-year-old woman comes to the physician because of a 3-month history of anxiety, weight loss and sensation of feeling hot even in cold weather. Her pulse is 118/min, blood pressure is 128/78 mm Hg. Physical examination shows sweaty palms and fine tremors of hands. Laboratory study shows elevated serum TSH, T3 & T4 levels. Which of the following is most likely the physiologic action of the thyroid hormones?

- A. Decreases cardiac contractility
- B. Decreases hepatic gluconeogenesis
- C. Decreases activity of the Na⁺-K⁺-ATPase in most cells
- D. Decreases lipolysis in the adipose tissue
- E. Increases protein synthesis

7. A 25-year-old woman is brought to the physician because of a 4-week history of palpitations and dizziness. Her blood pressure is 148/74 mm Hg, pulse is 122/min. Physical examination shows exophthalmos, restlessness, and fine tremors of hands. She is diagnosed with Graves' disease. A decrease in which of the following is most likely in this patient?

- A. TSH in plasma
- B. TSH-R Ab [stim] in plasma
- C. Serum free T3
- D. Thyroid gland size
- E. Basal metabolic rate

8. A 45-year-old woman is diagnosed with an autoimmune disease that results in hypersecretion from the endocrine gland shown in the figure below. She has not yet received any treatment.



Which of the following combinations of laboratory finding and clinical features is most likely in this patient?

Option	Serum TSH	Serum Glucose	Clinical feature
A	↓	↑	Lethargy
B	↑	↑	Weight gain
C	↓	↓	Increased body temperature
D	↓	↑	Proptosis
E	↑	↓	Non-pitting edema

9. A 35-year-old woman comes to the physician because of a 6-week history of palpitations, nervousness, sweating and a 4-kg (8.8-lb) weight loss. Her temperature is 38°C (100.4°F), pulse is 114/min and blood pressure is 142/70 mm Hg. Physical examination shows fine tremors of hands, lid lag and stare look, and a diffusely enlarged non-tender thyroid. Laboratory studies show a low serum concentration of thyroid-stimulating hormone (TSH) and an increased free T3 and T4. Increase in which of the following is most likely to increase heart rate in this patient?

- A. Contractility
- B. Heat production

- C. Beta receptor sensitivity
- D. Basal metabolic rate
- E. Vasodilation of skin blood vessels

10. A 32-year-old woman comes to the physician for a follow-up examination. She says she has lethargy, hair loss, and gained 5-kg (11-lb) weight during the last 6 weeks. She was diagnosed with Hashimoto's thyroiditis 3 years ago and is not adherent to thyroid supplements. Her temperature is 36.4C (97.5F), pulse is 52/min, and blood pressure is 132/88 mm Hg. Physical examination shows cold and dry skin, hair loss on body and scalp, puffiness of face, and non-pitting edema of the extremities. Serum studies show low T3, T4 concentrations but elevated TSH and TRH concentrations. Which of the following thyroid disorders is most likely in this patient?

- A. Primary hyperthyroidism
- B. Secondary hyperthyroidism
- C. Primary hypothyroidism
- D. Secondary hypothyroidism
- E. Iodine deficiency hypothyroidism

ANSWER KEY AND EXPLANATION

1. A 28-year-old man has decided to fast for 48 hours for charity. He is concerned and asks you if he needs to worry about the lack of iodine in his diet during this time causing decreased synthesis of thyroid hormones. Which of the following statements would apply in this case?

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- C. No deficiency as there is large iodine store in your bones
- D. There is deficiency as the liver breaks down all iodine
- E. No deficiency as they have a large iodine store bound to plasma proteins

Answer Key: A.

There is a large store of hormones in the form of colloid in the thyroid gland. Deficiency of iodine in diet must be over a longer interval of about 2 to 3 months before an individual would get deficiency of the hormone.

2. A 21-year-old woman comes the physician because of a 3-month history of 4.53-kg (10-lb) weight gain and difficulty coping with the winter weather. He says that he always feels cold even on a hot day. Physical examination shows puffy face and a 1-cm size nodule palpable in her neck. Blood studies show:

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Which of the following best describes this patient's condition?

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- B. Secondary hypothyroidism
- C. Primary hyperthyroidism
- D. Iodine deficiency
- E. Secondary hyperthyroidism

Answer Key: D.

This patient has hypothyroidism (Suggested by history, physical exam and low T4 levels). The patient TSH levels are high, hence it is not secondary hypothyroidism. Stimulatory TSH-R-AB is seen in Grave's disease as a cause of hyperthyroidism.

3. Under normal circumstances, which of the following structures concentrates the highest amount of iodine?

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- D. Liver
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Answer Key: C.

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- E. Activating tyrosine kinase

Answer Key: A.

TRH binds membrane receptors coupled to phospholipase C (PLC) by G proteins (Gq). PLC hydrolyzes plasma membrane phosphatidylinositol bisphosphate (PIP2),

generating inositol trisphosphate (IP3) and diacylglycerol (DAG). IP3 mobilizes intracellular stores of Ca^{2+} . The rise in Ca^{2+} stimulates thyroid-stimulating hormone (TSH) secretion. Ca^{2+} and DAG also activate protein kinase C (PKC), which stimulates gene expression for the α and β subunits of TSH.

5. A 34-year-old woman comes to the physician because of a 6-week history of heat intolerance, nervousness, increased appetite, and weight loss. Physical examination shows warm and moist skin, stare look, fine tremors of hands, and hyperreflexia. Laboratory studies show elevated serum concentrations of T3 and T4. Which of the following is the most likely mechanism of action of these hormones in the target cells?

- A. The IP3 and DAG second messenger system
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- C. Binding to receptors on the plasma membrane of the cell
- D. Binding to receptors in the nucleus of the cell
- E. Activating tyrosine kinase

Answer Key: D.

Thyroid hormones binds to the thyroid hormone receptors located in the nucleus of the target cells and turns on gene expression in the cells.

6. A 25-year-old woman comes to the physician because of a 3-month history of anxiety, weight loss and sensation of feeling hot even in cold weather. Her pulse is 118/min, blood pressure is 128/78 mm Hg. Physical examination shows sweaty palms and fine tremors of hands. Laboratory study shows elevated serum TSH, T3 & T4 levels. Which of the following is most likely the physiologic action of the thyroid hormones?

- A. Decreases cardiac contractility
- B. Decreases hepatic gluconeogenesis
- C. Decreases activity of the $\text{Na}^{+}\text{-K}^{+}\text{-ATPase}$ in most cells
- D. Decreases lipolysis in the adipose tissue
- E. Increases protein synthesis

Answer Key: E.

Thyroid hormone increases gluconeogenesis, cardiac contractility, lipolysis in adipose tissue and increases expression of $\text{Na}^{+}\text{-K}^{+}\text{-ATPase}$ in most cells. Thyroid hormones also promote protein synthesis. However, thyroid hormones, in excess, produce catabolic effect on proteins.

7. A 25-year-old woman is brought to the physician because of a 4-week history of palpitations and dizziness. Her blood pressure is 148/74 mm Hg, pulse is 122/min. Physical examination shows exophthalmos, restlessness, and fine tremors of hands.

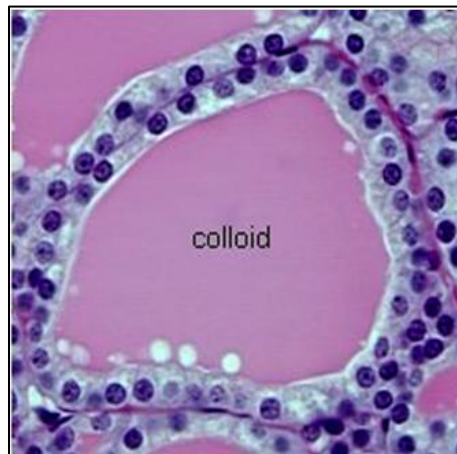
She is diagnosed with Graves' disease. A decrease in which of the following is most likely in this patient?

- A. TSH in plasma
- B. TSH-R Ab [stim] in plasma
- C. Serum free T3
- D. Thyroid gland size
- E. Basal metabolic rate

Answer Key: A.

In Graves' disease, there is increased levels of T3, TSH-R (stim) Ab, BMR & gland size. The elevated levels of T3 suppresses TSH by negative feedback.

8. A 45-year-old woman is diagnosed with an autoimmune disease that results in hypersecretion from the endocrine gland shown in the figure below. She has not yet received any treatment.



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E	↑	↓	Non-pitting edema

Answer Key: Option D. This patient has Graves' disease, an autoimmune disorder of thyroid. Note that the thyroid follicular cells are large cuboidal with hypersecretion of T3, T4. High T3,4 inhibit TSH secretion. Hyperthyroidism elevates serum glucose concentration. Protrusion of eyeballs (proptosis) is a prominent clinical feature in these patients.

9. A 35-year-old woman comes to the physician because of a 6-week history of palpitations, nervousness, sweating and a 4-kg (8.8-lb) weight loss. Her temperature is 38°C (100.4°F), pulse is 114/min and blood pressure is 142/70 mm Hg. Physical examination shows fine tremors of hands, lid lag and stare look, and a diffusely enlarged non-tender thyroid. Laboratory studies show a low serum concentration of thyroid-stimulating hormone (TSH) and an increased free T3 and T4. Increase in which of the following is most likely to increase heart rate in this patient?

- A. Contractility
- B. Heat production
- C. Beta receptor sensitivity
- D. Basal metabolic rate
- E. Vasodilation of skin blood vessels

Answer: C. Beta receptor sensitivity. Thyroid hormones increase the beta receptor sensitivity in heart. This results in increase in heart rate through SA node and increase in contractility through ventricle muscles. There is increased catecholamine secretion, and beta-adrenergic receptors are also upregulated in various tissues.

10. A 32-year-old woman comes to the physician for a follow-up examination. She says she has lethargy, hair loss, and gained 5-kg (11-lb) weight during the last 6 weeks. She was diagnosed with Hashimoto's thyroiditis 3 years ago and is not adherent to thyroid supplements. Her temperature is 36.4°C (97.5°F), pulse is 52/min, and blood pressure is 132/88 mm Hg. Physical examination shows cold and dry skin, hair loss on body and scalp, puffiness of face, and non-pitting edema of the extremities. Serum studies show low T3, T4 concentrations but elevated TSH and TRH concentrations. Which of the following thyroid disorders is most likely in this patient?

- A. Primary hyperthyroidism
- B. Secondary hyperthyroidism
- C. Primary hypothyroidism
- D. Secondary hypothyroidism
- E. Iodine deficiency hypothyroidism

Answer: C. Primary hypothyroidism. Hashimoto's thyroiditis is an autoimmune thyroid disease in which thyroid gland is destroyed by autoantibodies. This leads to reduced functioning of the gland and results in primary hypothyroidism. Laboratory shows low T3, T4 but elevated TSH and TRH. Because of a reduced negative feedback suppression on anterior pituitary and hypothalamus, the serum TSH and TRH increase.